

LAB EXPERIMENT RECORD

SET 6 Product Sales Analysis

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
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Register Number	192424049
Date	03/10/2026
Day	Saturday

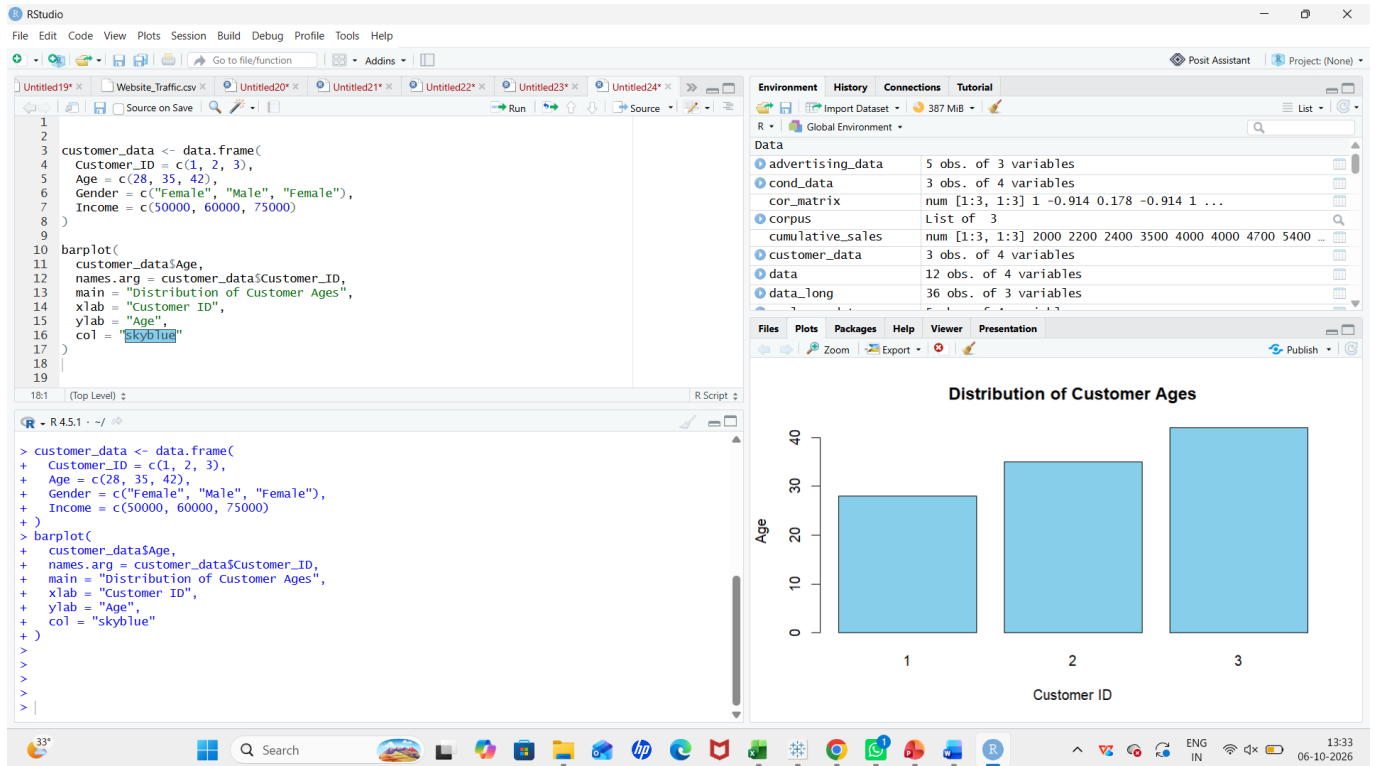
AIM

To analyze customer demographic data using R programming by creating a bar chart, pie chart, and table, and to develop a Tableau dashboard for interactive exploration of customer age, gender, and income information.

Customer ID	Age	Gender	Income (in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

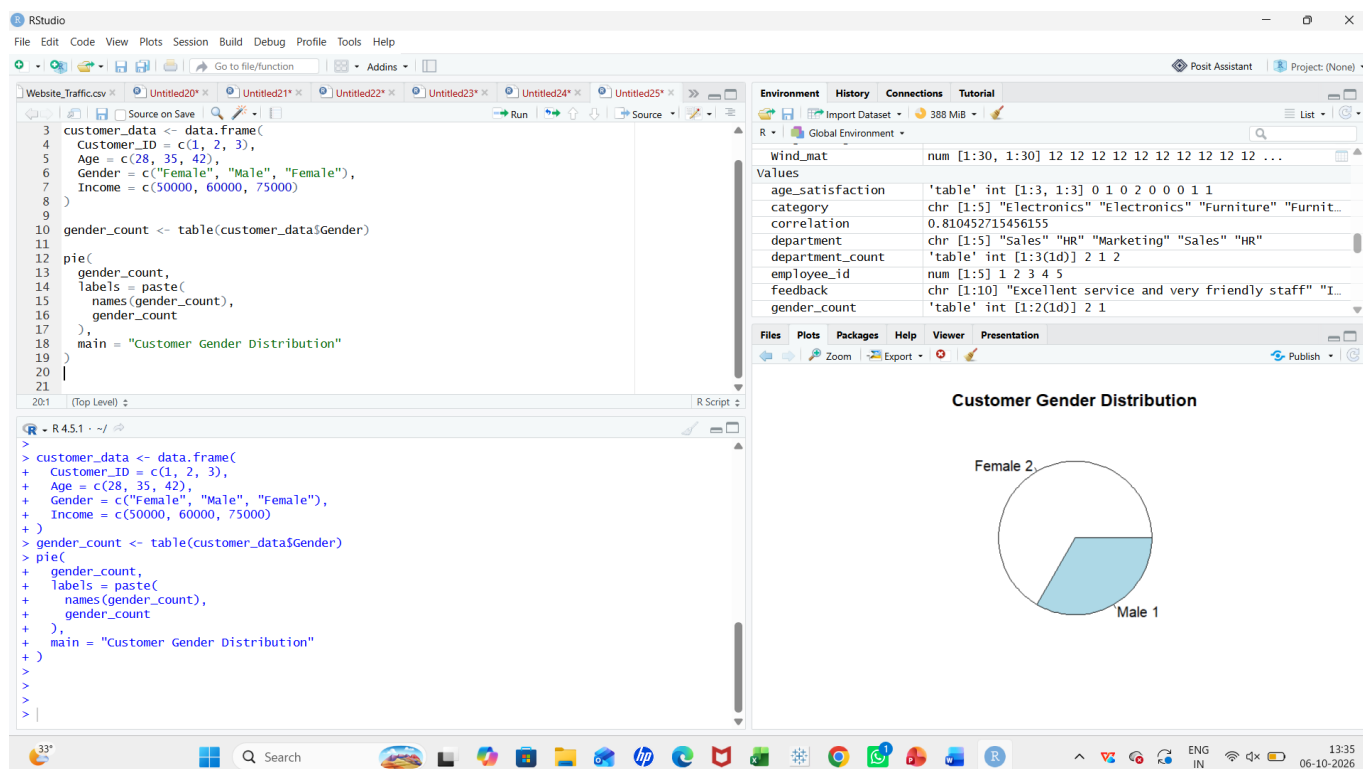
1. Create a bar chart to visualize the distribution of customer ages.

```
customer_data <- data.frame(  
  Customer_ID = c(1, 2, 3),  
  Age = c(28, 35, 42),  
  Gender = c("Female", "Male", "Female"),  
  Income = c(50000, 60000, 75000)  
)  
  
barplot(  
  customer_data$Age,  
  names.arg = customer_data$Customer_ID,  
  main = "Distribution of Customer Ages",  
  xlab = "Customer ID",  
  ylab = "Age",  
  col = "skyblue"  
)
```



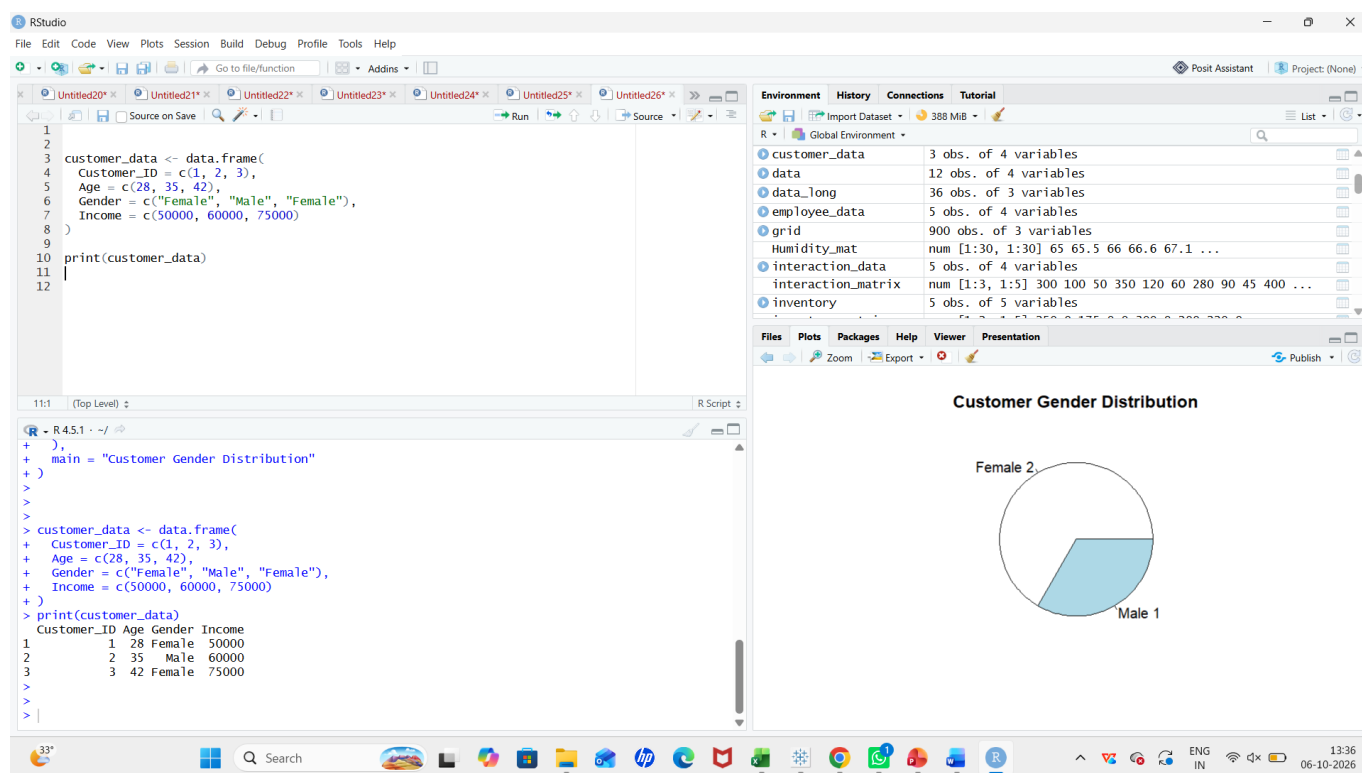
2. Generate a pie chart to represent the distribution of customers by gender.

```
customer_data <- data.frame(  
  Customer_ID = c(1, 2, 3),  
  Age = c(28, 35, 42),  
  Gender = c("Female", "Male", "Female"),  
  Income = c(50000, 60000, 75000)  
)  
  
gender_count <- table(customer_data$Gender)  
  
pie(  
  gender_count,  
  labels = paste(  
    names(gender_count),  
    gender_count  
  ),  
  main = "Customer Gender Distribution"  
)
```

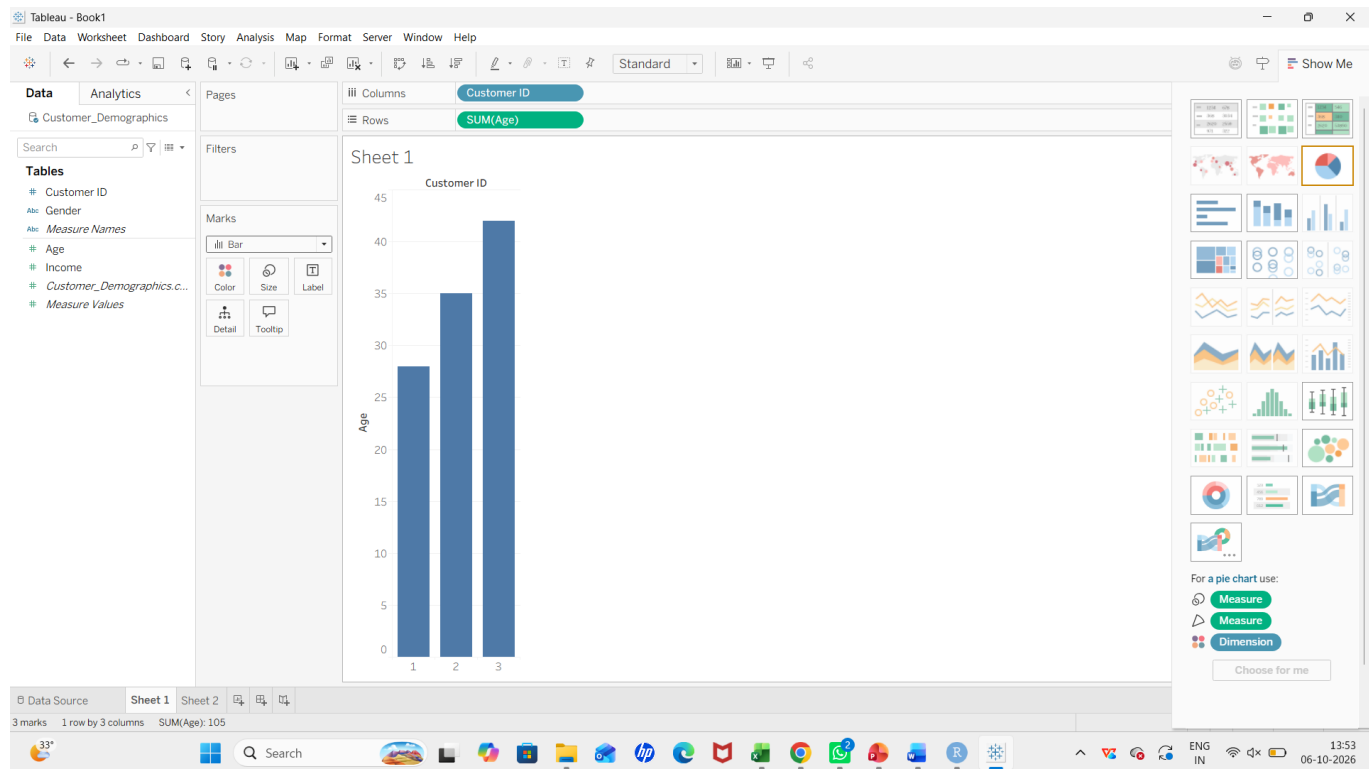


3. Build a table to show the demographic information for each customer.

```
customer_data <- data.frame(  
  Customer_ID = c(1, 2, 3),  
  Age = c(28, 35, 42),  
  Gender = c("Female", "Male", "Female"),  
  Income = c(50000, 60000, 75000)  
)  
  
print(customer_data)
```



14.



PROCEDURE

1. Create the Customer Demographics dataset in R with Customer ID, Age, Gender, and Income.
2. Use the `barplot()` function to create a bar chart showing the age of each customer.
3. Add appropriate axis labels and title to the bar chart.
4. Calculate the number of customers in each gender category.
5. Use the `pie()` function to create a pie chart showing the distribution of customers by gender.
6. Add suitable labels and title to the pie chart.
7. Create a table displaying the Customer ID, Age, Gender, and Income for each customer.
8. Import the customer demographic dataset into Tableau.
9. Create a bar chart in Tableau to display customer ages.
10. Create a pie chart in Tableau to show the gender distribution.
11. Create a table in Tableau to display the complete demographic information.
12. Combine the bar chart, pie chart, and table into a Tableau dashboard.
13. Add filters to allow users to interactively explore the customer demographic data.

RESULT

The customer demographic data was successfully analyzed using R programming. The bar chart displayed the age distribution of customers, while the pie chart showed the distribution of customers by gender. A table was created to display the complete demographic information, including customer ID, age, gender, and income. Finally, a Tableau dashboard was developed by combining the bar chart, pie chart, and table for interactive exploration of customer demographics.

